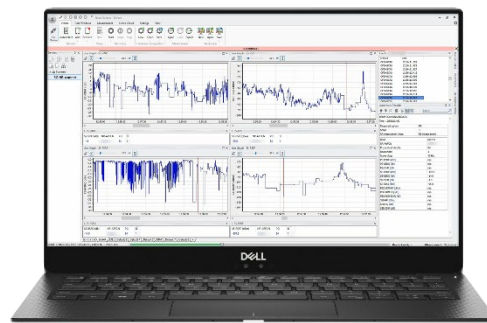


Nemo Outdoor

Powerful Drive Test Tool for Measuring and Monitoring Wireless Networks



Nemo Outdoor is a laptop-based, drive and laboratory testing solution for wireless networks, widely recognized as an unparalleled measurement and optimization tool for perfecting the air interface. It serves as a comprehensive solution for device testing, troubleshooting, optimization, and benchmarking throughout the entire network lifecycle.

The solution supports all the latest network technologies and flagship smartphones, including 5G NR (Non-Standalone, Standalone, and Carrier Aggregation), LTE-A, NB-IoT, LTE-M, and legacy 2G-3G mobile networks. Additionally, Nemo Outdoor provides explicit support for advanced functionalities such as NR Dual Connectivity (NR-DC).

For large-scale operations, Nemo Outdoor boasts support for over 340 different terminals, IoT devices, and scanning receivers from various vendors. The system is highly scalable, enabling the simultaneous use of up to 60 test devices to efficiently perfect the air interface of wireless networks.

The tool collects precise measurement results and geographical coordinates, providing crucial data for network planning, roll-out, tuning, verification, optimization, maintenance, and benchmarking purposes. Nemo Outdoor distinguishes itself by measuring Quality of Experience (QoE) metrics for services using the same native applications that real customers use (e.g., advanced YouTube video testing, Facebook, WhatsApp, Telegram, Signal, Dropbox, Viber, BiP, and Line), thereby reflecting the true end-user experience.

Nemo Outdoor maintains its status as an industry leader, always among the first-to-market with support for the very latest technologies, including terminals based on cutting-edge chipsets such as Qualcomm X85, X80, X75, X71, X62, and X35 (RedCap), Samsung Exynos 5400, 5328, 5153, and 2600, and MediaTek M90 and M60 modem variants.

Features

- Quality-of-experience (QoE) metrics for the services and applications your customers are actually using.
- Automated measurements with extensive scripts and large-scale measurement lists enable you to focus on the actual task at hand during drive testing.
- User-defined parameters from signaling messages can be searched and displayed in info view and graph side panel during measurement and playback.
- Versatile benchmarking capabilities, including Nemo Network Benchmarking Solution (NBM) and Nemo Backpack Pro.
- Added support for New Radio Dual Connectivity (NR-DC).
- Automatic BTS detection functionality.
- Dual Sim support.
- Mediatek Linux CPE device support.
- ViSQOL algorithm support for voice quality measurements.
- Wide portfolio of OTT applications with Nemo Application test automation: look at the complete list from your latest ATA application note.
- NTN scanning support.
- ITU-T P.1204 aligned Video QoE solution.

5G Features

- Supports Qualcomm (X35 RedCap, X50, X55, X60, X65, X70, X75, X85), Samsung Exynos (5100, 5123, 5123A, 5123B, 5133, 5153, 5300, 5400), and MediaTek (M60 RedCap, M70, M80, M90) modem-based devices, and 3rd party scanning receivers.
- Supports 5G NR standalone (SA) and non-standalone (NSA) mode testing.
- Supports 5G NR carrier aggregation testing.
- Supports 5G NR NSA/SA forcing.
- Supports 5G NR channel/PCI forcing.
- Collects 5G NR beam specific KPIs.
- Supports massive MIMO (mMIMO) drive tests and field measurements.
- Supports Dynamic Spectrum Sharing (DSS) measurements.

Reduce the Time-to-Market of Your New Services

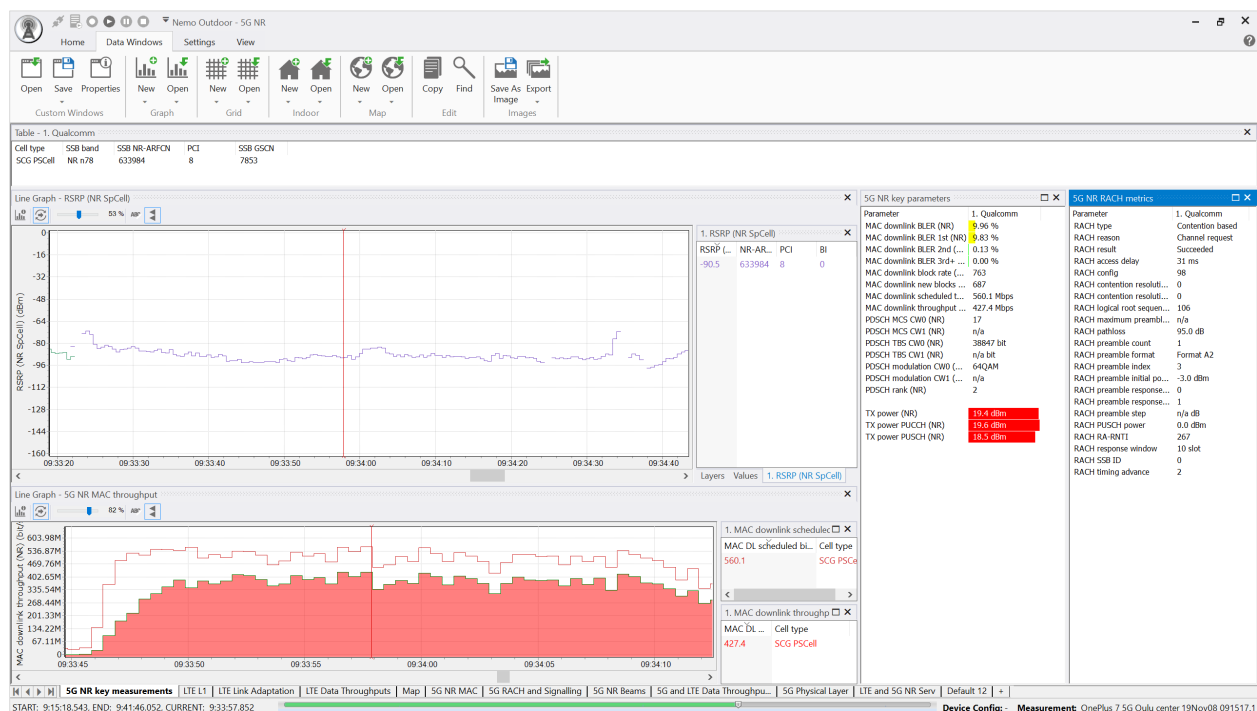
To gain competitive advantage in today's market, monitoring and improving customer experience is crucial. The information provided by Nemo Outdoor assists in the verification and troubleshooting of new services by reducing the time-to-market.

Nemo Outdoor's diversity is only matched by its flexible user interface and ease of use. Owing to its single platform, all essential functionalities can be found in one location. Through the fully customizable user interface, the user is able to tailor Nemo Outdoor to their particular use.

5G Ready

Nemo Outdoor supports New Radio Dual Connectivity (NR-DC), real 5G NR field measurements, both standalone (SA) and non-standalone (NSA) modes, and 5G NR forcing features, and offers the widest range of 5G NR KPIs on the market. The solution combines Nemo Outdoor with Qualcomm (X35 RedCap, X50, X55, X60, X65, X70, X75, X85), Samsung Exynos (5100, 5123, 5123A, 5123B, 5133, 5153, 5300, 5400), and MediaTek (M60 RedCap, M70, M80, M90) modem-based devices and 3rd party scanning receivers. The collected metrics include RACH information, TX power, rank (MIMO mode), modulation, MAC throughput and BLER, signal strength, and quality metrics of the SSB beams. In addition, we collect QoS measurements including throughputs from different layers and latency.

The 5G NR test system can be used to perform real 5G NR measurements with demodulation of the 5G NR reference signals and it enables simultaneous 2G, 3G, 4G, and 5G NR spectrum scan and frequency scan (CW) measurements. When paired with Nemo Network Benchmarking Solution (NBM), Nemo Outdoor offers a revolutionary 5G benchmarking solution.



Comprehensive Solution Based on a Single Laptop

Nemo Outdoor allows operators to test and verify the capacity of wireless networks and it comes with an incomparable support for over 300 test terminals and scanning receivers.

Nemo Outdoor is highly suitable for targeted network problems in all the established and emerging phases of the network life cycle. It can be expanded to cover extensive and powerful measurements from drive testing to benchmarking (indoors and outdoors) and QoS measurements. All this on a single laptop-based software platform.

Nemo Outdoor enables the measuring of Quality-of-Experience (QoE) metrics for the services and applications your customers are actually using. With the automated AI-driven Application Testing Automation method, mobile operators can reliably and rapidly optimize 5G network performance and deliver greater Quality of Experience (QoE) for smartphone users accessing world's most widely used Over-the-Top (OTT) services and social media applications.

In addition to drive test and benchmarking measurements, the Nemo Outdoor platform is expandable to support Dual Sim, Automatic BTS detection functionality, NTN scanning, voice (POLQA/POLQA3 and ViSQOL) and video quality (ITU-T P.1204) measurements and Keysight's Streaming Experience Quality Inspector, an algorithm for measuring real consumer QoE for YouTube streaming. Nemo Outdoor also offers tools for various troubleshooting and verification tasks, such as DAS anomaly analysis, real-time RF ingress and missing neighbor detection, pilot pollution analysis, GSM interference analysis, and Qualcomm and Samsung diagnostic data logging. In addition, with the advanced mobile band scan feature, Nemo Outdoor continuously scans for pilot channels from specified bands and automatically adds them for pilot scanning.

The optional Nemo Active Test Application (NATA), Keysight's proprietary communications interface and application, enables simple and productive deployment of smartphones in data benchmarking and voice quality measurements by utilizing the smartphones' independent capacity more effectively. With NATA, Nemo Outdoor can be used to perform testing with non-rooted devices as well.

Keysight enables innovators to push the boundaries of engineering by quickly solving design, emulation, and test challenges to create the best product experiences. Start your innovation journey at www.keysight.com.



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